

## **Re-evaluation of Prediction for Mashable News Popularity**

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### **Abstract**

The study revisits Fernandes et al. 's research in 2015 on Mashable News Popularity Prediction. Using their dataset with 39644 observations from UCI Machine Learning Repository, I analyzed 59 feature variables to predict number of shares of an article. By implementing various statistical learning methods, I found that while models show similar predictive performance, the gradient boosting method yields slightly lower error and more stable residual patterns, and I successfully replicated Fernandes et al. 's random forest classification results. Validating backward with a regression-based random forest model, I showed that the classification done in their study is proven to be correct. More generally, the study implies that popularity of online news articles is only moderately predictable with existing predictors, but it provides a standard procedure for social science problems of forecasting target variable with multivariate features.

# Re-evaluation of Prediction for Mashable News Popularity

## 1 Introduction

As modern society gains increasing attention by online platforms, the popularity of online news articles is often shown by shares. To analyze the various factors that may contribute to an article's popularity, Kelwin Fernandes et al. (2015) studied data from Mashable and performed both prediction and optimization to find out how authors could potentially benefit from manipulable adjustments to articles that help to increase popularity. Among groups of features they extracted from Mashable over a two-year span (including keyword shares, weekday/weekend, length, polarity, sentiment rates, title subjectivity, self-reference shares, lexical rates, meta elements, global subjectivity), researchers adopted an Intelligent Decision Support System (IDSS) with Artificial Business Intelligence (ABI) and found that under Random Forest modelling the keyword based-features and natural language processing features are the most important ones. In their study, they tried to classify the popularity into binary response defining shares  $D_1 > 1400$  as popular. Based on this prerequisite the model they prefer is random forest (Fernandes et al., 2015).

Their results are promising and insightful in interpreting how these features contribute to the post popularity of these articles. By dividing *shares* manually into unpopular & popular, they reached a plausible conclusion given that the final prediction model gives approximately 0.7 accuracy and precision. However, a possible reevaluation of the study would question the selection of threshold on popularity,  $D_1$ . As said in the study, it is based on an overall balanced consideration from *shares* variable with no other qualification. That could possibly lead to higher noise in the model and reduction in actual variance. The chance of false positive will then arise.

Although their study still provides a reasonable explanation of popularity level given a specific threshold, to address the potential question, in my following study I will further explore the same dataset and employ both regression method (for continuous response variable) and similar classification method. The purposes of my study are 1) to find and compare models to predict *shares* with regression analysis; 2) to reexamine and confirm the existing conclusion of Kelwin Fernandes et al. Using *shares* as the target variable and same explanatory variables as they did, I will use random forest with different thresholds, ordinary linear squares with converted  $\log(\text{shares} + 1)$ , shrinkage methods (ridge, elastic net), principal components regression (PCR), and tree-based methods in the following sections. I believe by summarizing the findings of multiple approaches we will obtain a more comprehensive understanding of the factors driving popularity of a news article posted online on Mashable.

## 2 Methods

I first employed ordinary linear squares regression to fit the data. Since the dataset has large size ( $n = 39644$ ), OLS provides a quick and computationally convenient overview of the problem. However, the response variable, *shares*, is skewed with extreme outliers, I modeled  $\log(\text{shares} + 1)$  instead of the raw counts to mitigate the effect of the outliers. The preliminary diagnostics (Figure 1.1) of OLS showed heteroskedasticity and non-linearity with a low  $R^2$ . Based on the above evaluation, though many parameters were statistically significant, the p-values are unstable due to potential collinearity ( $\text{VIFs} > 10$ , Figure 1.2). Thus, I need to adopt methods that are more robust to these effects and adjust the parameters on my end to possibly match the results by Fernandes et al.'s.

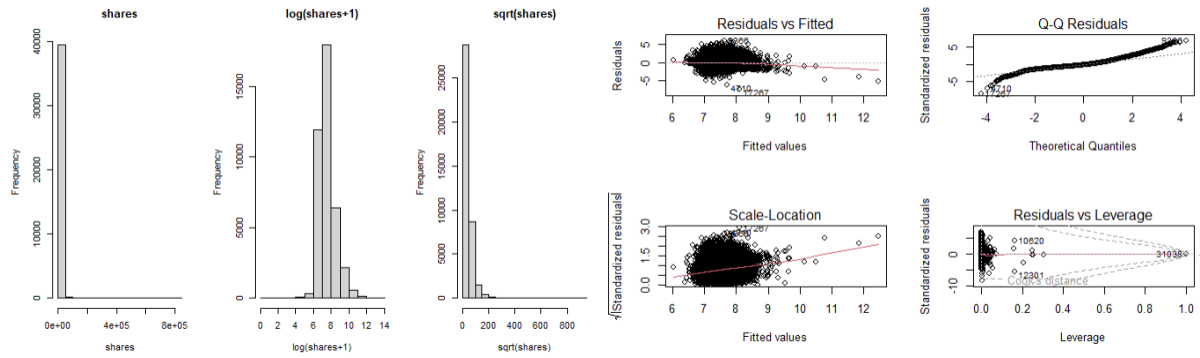


Figure 1.1

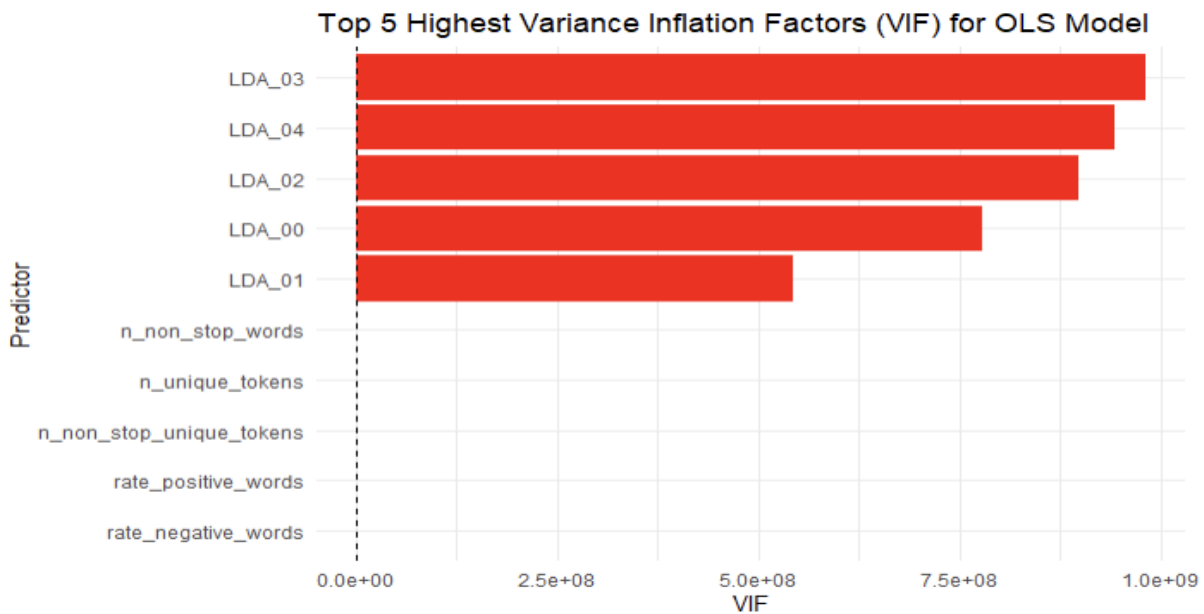


Figure 1.2

## 2.1 Shrinkage

### 2.1.1 Ridge

Ridge regression changes the ordinary least squares objective by add an  $\ell_2$  penalty on the coefficients (Hoerl & Kennard, 1970). In our case, it makes ridge regression a viable candidate to deal with correlated predictors with prediction accuracy. Based on 10-fold cross-validation with `cv.glmnet` in R, I obtained the ridge regression model using 70% of the original data as the

training dataset and the remaining as the validation set. The test RMSE gives approximately 0.87, showing reasonably strong predictive performance compared to the OLS baseline while explicitly accounting for multicollinearity through shrinkage.

### 2.1.2 Elastic Net

Like the ridge method, elastic net regression tackles with the collinearity problem with penalty term  $\ell_1$  and  $\ell_2$ . Since we do not know the exact functioning features yet, we generally do not want to drop variables directly as in lasso regression. To keep interpretability and predictive power, we employed elastic net here. The test RMSE gives approximately 0.87, like the ridge regression one, except smaller by a few thousandths.

The two shrinkage methods both hold exceptionally low test RMSE and good prediction. Comparably, they have very similar results in the sense of predicting popularity of Mashable news.

Figure 2.1 (a)

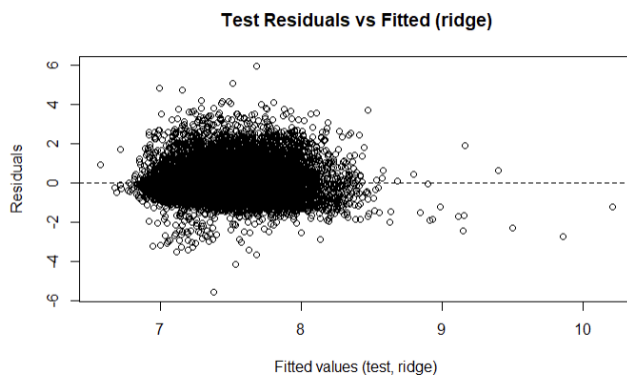
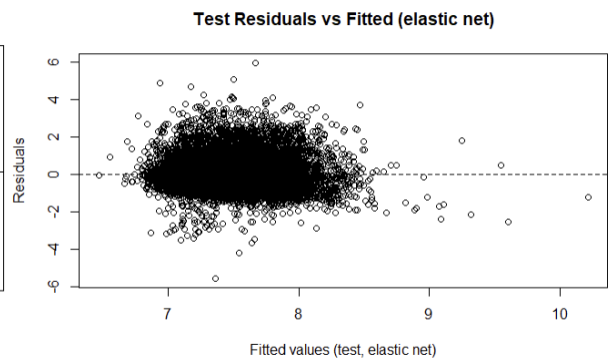


Figure 2.1 (b)



## 2.2 Principal Component Regression (PCR)

Another prominent method in coping with multicollinearity and copious amounts of predictors is Principal Component Regression (PCR), which reduces the dimensionality of the original data.

In practice, first, using the same training set as in shrinkage methods, I performed principal component analysis (PCA) on the standardized predictor matrix  $X$ . Then with cross-validation on  $K = 51$  I fitted the final PCR model and evaluated it on the held-out test data to obtain test  $\text{RMSE} \approx 0.87$ .

According to the analysis, the first principal component appears to focus on the sentiment-driven index. Higher values of PC1 correspond to shorter, more negative World news with strong keywords. In the PCR model, the coefficient for PC1 is positive, indicating that moving toward the “more negative World news with strong keywords direction” is associated with higher predicted popularity, after accounting for the remaining components.

### 2.3 Tree-Based Methods

Finally, for the regression study, I adopted boosted regression tree model using `gbm` in R. At each boosting iteration, a new tree fits the current residuals, so the ensemble gradually corrects errors itself. With the same train-validation procedure as above, the GBM model with a Gaussian loss gives a  $\text{RMSE} \approx 0.86$ .

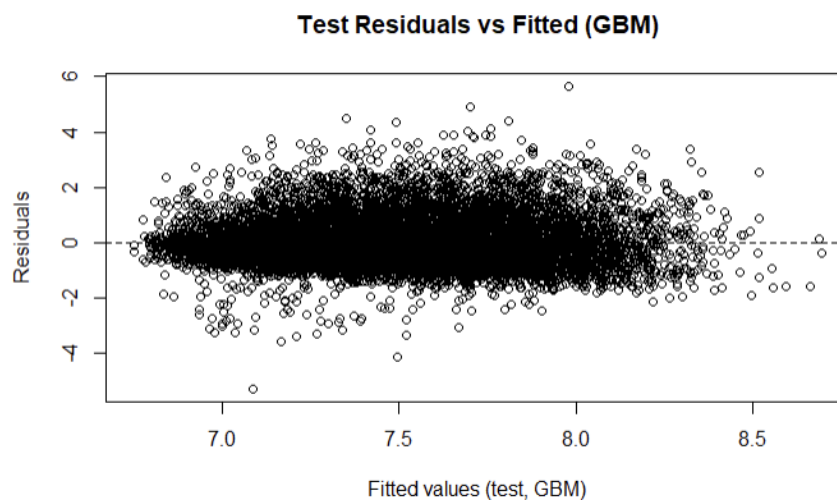


Figure 2.2

### 2.4 Validation of Random Forest Modelling

In the prediction method module, I lastly validated the modelling in Fernandes et al.'s study. Using the same  $D_1 = 1400$  and dividing the *shares* variable into unpopular / popular categories, I fitted the random forest model with the default S3 method. Then I compared the metrics with their results. Since unknown parameters of random forest are used in their study and different random seeds may be applied, the results do not match overall as in numerical value. However, the deviation is small for accuracy and AUC, both differ only by hundredths (Figure 3.1.1). Additionally, I arranged the importance of predictors by Gini (Figure 3.1.2). Unfortunately, the top ten most important predictors are not the same across mine and theirs. This could be expected because the results depend on the specific data split, and details of feature construction. Given that, I focused on the agreement at the level of feature groups (keyword statistics, topic closeness, self-reference shares, and subjectivity) rather than exact variable ranks. In that case, the results align with the previous anticipation.

Random Forest classification performance

| Model              | Accuracy | Precision | Recall | F1   | AUC  |
|--------------------|----------|-----------|--------|------|------|
| Random Forest (RF) | 0.66     | 0.66      | 0.66   | 0.66 | 0.72 |

Figure 3.1.1: Model Metrics for Random Forest with  $D_1 = 1400$

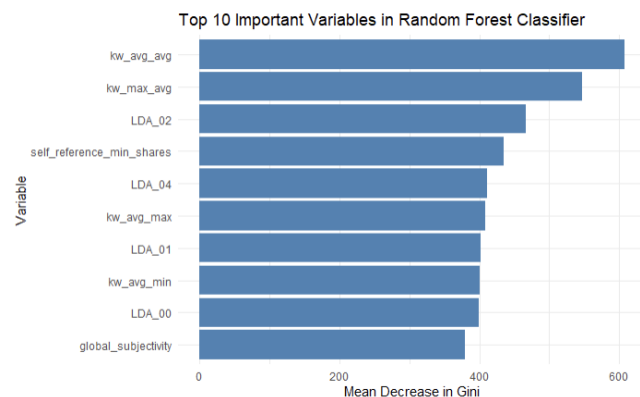




Figure 3.1.2: Top Ten Important Variables in RF Classifier by Gini

Later I performed a regression analysis based on random forest. I then categorized the regressed results into unpopular / popular variables using  $D_1 = 1400$ . Lastly, a comparison between the two methods of random forest is processed (Figure 3.2). The test confusion matrix shows that out of 11,894 articles, 5,095 tremendously popular articles were correctly identified as popular and 762 were missed (FN), while 2,284 unpopular articles were correctly classified and 3,753 were incorrectly labeled as popular (FP).

Comparison of Random Forest Classification Approaches

| Model                              | Accuracy | Precision | Recall | F1   | AUC  |
|------------------------------------|----------|-----------|--------|------|------|
| RF Classification                  | 0.66     | 0.66      | 0.66   | 0.66 | 0.72 |
| RF Regression-based Classification | 0.62     | 0.58      | 0.87   | 0.69 | 0.71 |

Figure 3.2

### 3 Results

Due to the large sample size for both training and test dataset, the shrinkage methods, PCR, GBM and random forest lead to a remarkably similar predictive power on the popularity of Mashable news (Figure 4). However, a possible tradeoff could exist between the former three methods and the GBM method. Based on the residual plot of fitted values, GBM forms an even trend across the whole fitted range, and variance is relatively more constant compared to the other three modellings; while those models give us better interpretability, especially the PCA I have done revealing the groupings of several features. To conclude, GBM might be a better, reliable but possible conservative one for us to predict the future Mashable News popularity.

### 4 Discussion

Using OLS, Shrinkage (ridge & elastic net), PCR, GBM, and Random Forest, all the methods have a quite similar predictive power in terms of RMSE (Figure 4). Out of the six, the GBM holds the most stable fitted residual, showing a more reliable performance on the real data. I would conclude that using GBM as the best prediction model for online news popularity. In Fernandes et al.'s study, as their final goal was to build a optimization system to revise a pre-post article, I believe both GBM and Random Forest will complete the task, with the price of losing interpretability. Despite that, we still gained a comprehensive perspective when modeling PCR, and we saw that negative attitude with keywords component in an article makes significant difference to popularity.

| Model            | RMSE | MAE  |
|------------------|------|------|
| OLS              | 0.87 | 0.65 |
| Ridge Regression | 0.87 | 0.65 |
| Elastic Net      | 0.87 | 0.65 |
| PCR              | 0.87 | 0.65 |
| GBM              | 0.86 | 0.64 |
| RF Regression    | 0.86 | 0.64 |

Figure 4

Successfully, my results aligned with Fernandes et al.'s under  $D_1 = 1400$ . We both have comparably similar modeling metrics. To explore the effect of label noise of that threshold, the latter regression random forest model I adopted first regressed on the *shares* variable, and then classified the target variable. One objective of the approach is that it keeps the same framework as other models, so that we can use a continuous scale via RMSE to compare them. Additionally,

by thresholding afterward, the model exploited information about how far each article is above or below the cutoff, offering us a more flexible clarification. In principle, this improved recall for truly popular articles and translated prediction to a real-world question.

Besides, there are limitations existing in the study. With only one dataset from Mashable implemented, the notion of popularity is restricted to a conditional computation. We could not expand the conclusion to general online news due to potential confounding factors. And our dataset is skewed, which led to the conservative approach I adopted during modeling. Secondly, even with log-transformed *shares*, I still see some heteroskedasticity, thus linear modeling assumptions may be poorly satisfied. Lastly, across all six regression/classification task, the same train-test sets are used.

Based on those points made, future work could span the current analysis to models with different  $D_1$  value, in order to see how lenient or strict threshold may affect the prediction. An overall goal would be to generalize the results. Data from multiple online platforms will certainly help us to reflect a multi-faceted nature of articles popularity. The study also presents us with a common challenge for various topics in social science –a complex dynamics from different aspects of contemporary information on the internet.

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